CREDIT RISK PREDICTION

Project Report

This project uses customer and loan details to train machine-learning models that predict the credit-risk class recorded in the dataset.

Dataset target: *credit_risk*

Number of records: 1,000 (300 in Class 0 and 700 in Class 1)

Note: The dataset's exact meaning for Class 0 and Class 1 was not provided. This report keeps the original class numbers and does not assume which one means good or bad credit risk.

Prepared using the dataset summaries and model results shared for this project.

1. Project Overview

The aim of this project is to predict the credit-risk class using details such as loan amount, loan duration, credit history, savings, employment duration, age, loan purpose, and other customer information. Different machine-learning models were trained and compared. XGBoost was also tuned, and H2O AutoML was tested in a separate notebook.

2. Dataset Details

The dataset contains 21 columns: 20 input columns and one target column, *credit_risk*. The input columns are status, duration, credit_history, purpose, amount, savings, employment_duration, installment_rate, personal_status_sex, other_debtors, present_residence, property, age, other_installment_plans, housing, number_credits, job, people_liable, telephone, and foreign_worker.

Target class	Number of records	Percentage
Class 0	300	30%
Class 1	700	70%
Total	1,000	100%

The target is not evenly split. Class 1 has more than twice as many records as Class 0. Because of this, accuracy alone is not enough to judge a model.

3. How Credit Risk Is Analysed

The model learns from examples in the dataset. It looks for patterns between the input columns and the recorded credit_risk value. After training, it predicts a class for new records. Some models also provide a probability for each class.

Steps used to analyse risk:

1. Check the target column and count how many records belong to each class.

2. Compare the classes using loan amount, age, loan purpose, duration, credit history, savings, and other available columns.

3. Prepare the data for machine learning by handling missing values and converting text categories into values the models can use.

4. Train the models on training data and check their results on validation or test data.

5. Compare accuracy, precision, recall, F1 score, and ROC-AUC to understand different types of model errors.

6. Check which class is being missed or predicted incorrectly before deciding how the model could be used.

4. Findings from the Dataset

4.1 Loan amount by class

Class	Count	Mean amount	Median amount	Minimum	Maximum
0	300	3,938.13	2,574.5	433	18,424
1	700	2,985.46	2,244.0	250	15,857

Class 0 has a higher average and median loan amount than Class 1 in the summary provided.

4.2 Age by class

Class	Count	Mean age	Median age	Minimum	Maximum
0	300	33.96	31	19	74
1	700	36.22	34	19	75

Class 1 has a slightly higher average and median age than Class 0. Age alone does not explain the target class.

4.3 Loan purpose by class

Loan purpose	Class 0 (%)	Class 1 (%)
Business	11.11	88.89
Car (new)	38.03	61.97
Car (used)	16.50	83.50
Domestic appliances	22.14	77.86
Education	36.36	63.64
Furniture/equipment	41.67	58.33
Others	35.05	64.95
Radio/television	32.04	67.96
Repairs	33.33	66.67
Retraining	44.00	56.00

Class 1 is more common than Class 0 in every listed loan-purpose group. Retraining has the closest split, while business has the highest Class 1 share. These percentages describe this dataset and do not prove that loan purpose causes the outcome.

4.4 Loan amount groups

Amount group	Class 0 (%)	Class 1 (%)
0–2K	28.01	71.99
2K–5K	26.58	73.42
5K–10K	36.49	63.51
10K–20K	60.00	40.00

The Class 0 share is highest in the 10K–20K group. The class pattern changes across the amount groups, so loan amount may help the models separate the classes.

5. Kurtosis Results

Kurtosis describes how heavy or light the tails of a distribution are compared with a normal distribution. The values below are interpreted as excess kurtosis, as commonly returned by pandas.

Feature	Kurtosis	Simple reading
duration	0.920	Heavier tails than normal
amount	4.293	Much heavier tails; check for extreme amounts
installment_rate	-1.210	Lighter tails than normal
present_residence	-1.381	Lighter tails than normal
age	0.596	Slightly heavier tails
number_credits	1.604	Heavier tails than normal
people_liable	1.649	Heavier tails than normal
credit_risk	-1.238	Binary target; not useful for checking normality

The amount column has the highest kurtosis. This may mean that some loan amounts are far from the centre of the distribution. Kurtosis alone does not confirm outliers.

6. Machine-Learning Model Comparison

The following table uses the model comparison values supplied for the project. The original table headings appeared slightly shifted, so the metrics are shown in the order inferred from the values shared: accuracy, precision, recall, F1 score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1 score	ROC-AUC
Random Forest	0.780	0.7963	0.9214	0.8543	0.8037
Gradient Boosting	0.760	0.8151	0.8500	0.8322	0.7799
XGBoost	0.730	0.7905	0.8357	0.8125	0.7599
Logistic Regression	0.705	0.7755	0.8143	0.7944	0.7594
LightGBM	0.740	0.7933	0.8500	0.8207	0.7523
Decision Tree	0.685	0.7251	0.8857	0.7974	0.6799

Random Forest has the highest accuracy, F1 score, and ROC-AUC in this comparison. Gradient Boosting has the highest precision. The best model depends on which kind of mistake matters most for the project.

7. XGBoost Tuning and Test Results

The XGBoost model was tuned using cross-validation. The best settings found were:

Parameter	Selected value
n_estimators	300
max_depth	4
learning_rate	0.01
subsample	0.7
colsample_bytree	1.0

Best cross-validation ROC-AUC during tuning: 0.7959.

Test metric	Result
Accuracy	72.50%
Precision	77.07%
Recall	86.43%
F1 score	81.48%

Test metric	Result
ROC-AUC	77.25%

The tuned model's recall is higher than its accuracy. This means it found many of the actual positive-class cases, but some predictions were still incorrect. The exact meaning of a positive result depends on which target value was treated as positive.

8. H2O AutoML Results

H2O AutoML tested models automatically and reported results for a Stacked Ensemble. The training results and cross-validation results are kept separate below because they were measured in different ways.

8.1 Stacked Ensemble training results

Metric	Training result
MSE	0.0889
RMSE	0.2981
Log Loss	0.3040
AUC	0.9611
AUCPR	0.9824
Gini	0.9221
Null deviance	977.3829
Residual deviance	486.4124
AIC	496.4124

The training AUC is high, but it describes the model's results on training data. It should not be used alone to estimate performance on new records.

8.2 H2O AutoML five-fold cross-validation

Metric	Mean result
Accuracy	77.30%
AUC / ROC-AUC	0.7959
Precision	0.7734
Recall	0.9588
F1 score	0.8554
PR-AUC	0.8988
Specificity	0.3440
MSE	0.1607
RMSE	0.4006

Across the five validation folds, the model had high recall but low specificity. In simple terms, it identified many records from the positive class but also incorrectly marked some negative-class records as positive. The label mapping should be checked before deciding what this means for credit risk.

9. How Risk Can Be Categorised

The dataset records two classes, 0 and 1. The model can predict one of these classes. To describe applicants as lower, medium, or higher risk, first confirm which class means the bad credit outcome. Then use the model's probability for that outcome.

Probability of the bad outcome	Suggested category
Below 30%	Lower predicted risk
30% to below 60%	Medium predicted risk
60% or higher	Higher predicted risk

These ranges are examples to explain the idea. They were not calculated from the supplied dataset results and should not be used as final lending cut-offs without testing them on validation data and considering the cost of different mistakes.

Example: If the confirmed bad-outcome probability for a record is 0.72 (72%), the example bands would place it in the higher predicted-risk group. This is a model estimate, not a guarantee of repayment or default.

10. Conclusion

This project compared six machine-learning models and tested H2O AutoML for credit-risk classification. In the supplied model comparison, Random Forest had the strongest results across accuracy, F1 score, and ROC-AUC. Tuned XGBoost reached 72.5% test accuracy and 0.7725 ROC-AUC. H2O AutoML reported 77.3% mean cross-validation accuracy and 0.7959 AUC, with high recall but low specificity.

The results suggest that the models can find patterns in the dataset, but the final model should be chosen only after checking the class labels, comparing models on the same untouched test set, and reviewing the confusion matrix. The class numbers must be mapped to good/bad risk before the results are described as safe or risky lending outcomes.

11. Next Steps

- Confirm what Class 0 and Class 1 mean in the original dataset.
- Evaluate Random Forest and the H2O AutoML leader on the same test set.
- Review false positives and false negatives with a confusion matrix.
- Choose risk bands only after checking predicted probabilities on validation data.
- Keep a record of the preprocessing steps and model settings.

Report note: This report uses the statistics and model results provided for the project. It does not add new measurements or claim that the models have been independently rerun.